2.3.6 Wrap-Up: Ticket Out the Door

1. Find the value of each of the following.

   a. $\sqrt[3]{125} =$

   b. $\sqrt[3]{-8} =$

2. Solve each equation.

   a. $w^3 = -64$

   b. $1 = k^3$

Unit 2, Lesson 4: Approximating Irrational Numbers

Benchmarks

MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.

MA.8.NSO.1.2 Plot, order, and compare rational and irrational numbers represented in various forms.

Learning Targets

- ☐ I can identify whether a number is rational or irrational.
- ☐ I can approximate the values of irrational numbers.
- ☐ I can plot irrational numbers on a number line.

2.4.1 Warm-Up: Number Pattern

A pattern is shown with one value missing.

<table>
<tr><td colspan="3"></td><td colspan="2"></td><td colspan="3"></td></tr>
<tr><td colspan="2"></td><td colspan="2">25</td><td colspan="2">22</td><td colspan="2"></td></tr>
<tr><td></td><td colspan="2">12</td><td colspan="2">13</td><td colspan="2">9</td><td></td></tr>
<tr><td colspan="2">2</td><td colspan="2">10</td><td colspan="2">3</td><td colspan="2">6</td></tr>
</table>

1. Determine the value of the empty cell in the pattern.

2. Explain how you determined the missing value.

2.4.2 Collaboration: Square Roots vs. Cube Roots

Work with your partner to complete the following.

1. Damion knows that $8^2 = 64$ and $4^3 = 64$. He says this means the following equations are all true.

Equation A	$\sqrt{64} = 8$
Equation B	$\sqrt[3]{64} = 4$
Equation C	$\sqrt{-64} = -8$
Equation D	$\sqrt[3]{-64} = -4$

a. Discuss with your partner whether or not you agree that all of the equations Damion wrote are true.

b. If Damion wrote any false equation(s), explain what makes them false.
If all of Damion's equations are true, explain how you know.

2. Complete each statement.

a. $\sqrt[3]{-1}$ is
- ○ equal to ______
- ○ not a real number

because

- ○ ______ $\times$ ______ $\times$ ______ $= -1$.
- ○ ______ $\times$ ______ $= -1$.
- ○ no real number multiplied by itself results in -1.

b. $\sqrt{-1}$ is
- ○ equal to ______
- ○ not a real number

because

- ○ ______ $\times$ ______ $\times$ ______ $= -1$.
- ○ ______ $\times$ ______ $= -1$.
- ○ no real number multiplied by itself results in -1.

3. Work with your partner to complete the table by choosing a value of k based on the inequality. Then, determine if the result is real or non-real.

Description	**Example**	**Real or Non-Real Result(s)**
$\sqrt{k}$ where $k > 0$		○ real ○ non-real
$\sqrt{k}$ where $k < 0$		○ real ○ non-real
$\sqrt[3]{k}$ where $k > 0$		○ real ○ non-real
$\sqrt[3]{k}$ where $k < 0$		○ real ○ non-real

4. Write a short explanation of when a non-real result will occur.

2.4.3 Guided Instruction: Rational and Irrational Numbers

The real number system includes both ***rational*** and ***irrational numbers***.

A ***rational number*** is a real number that can be expressed as a ratio of two integers.

Rational numbers can be written as a fraction, $\frac{a}{b}$, where a and b are integers and $b \neq 0$, or as a decimal. In decimal form, rational numbers either terminate or repeat.

An ***irrational number*** is a real number that cannot be expressed as a ratio of two integers.

When expressed as decimals, irrational numbers do not terminate or repeat. Therefore, any decimal expansion is only an approximate value.

1. The relationship between the circumference and diameter of any circle is represented by pi (π).

   a. Discuss with your partner what you remember about the value of π.

   b. Complete the statements.

   The ratio of the circumference of a circle to its diameter is expressed as π. The values 3.14 and $\frac{22}{7}$ are

   ○ equivalent to, or the exact value of
   ○ used to approximate the value of

   π.

   The exact value of π

   ○ can
   ○ cannot

   be expressed as a fraction of two integers or a decimal. Therefore, π is

   ○ a rational
   ○ an irrational

   number.

2. Enter each of the following values into a calculator.

$\sqrt{25}$ $\sqrt{18}$ $\sqrt[3]{27}$ $\sqrt[3]{30}$

a. Circle the values that can be expressed as a ratio of two integers.

b. Discuss with your partner how you would describe the values that are not circled.

c. Complete the statements.

Square roots of perfect squares and cube roots of perfect cubes ○ can ○ cannot be expressed as a ratio of two integers. Therefore, they are ○ rational. ○ irrational.

Square roots of non-perfect squares and cube roots of non-perfect cubes ○ can ○ cannot be expressed as a ratio of two integers. Therefore, they are ○ rational. ○ irrational.

2.4.4 Exploration: Roots of Non-Perfect Squares and Non-Perfect Cubes

The decimal expansions of irrational numbers do not terminate nor repeat. Therefore, the values of a square root of a non-perfect square can only be approximated.

The ***principal square root*** is the positive square root of a positive real number.

When evaluating the square root of a number, the ***principal square root*** is assumed when the radical symbol is used. A negative symbol, $-$, is placed in front of the radical to indicate the negative square root. Perfect squares can be used to estimate the value of the principal square root of non-perfect squares.

1. Work with your partner to complete the following to approximate the principal square root of $\sqrt{14}$.

   a. Complete the inequality to show which two perfect squares 14 lies between.

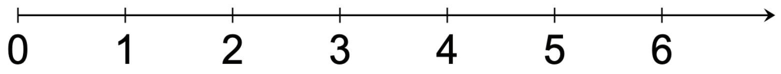
   $$\_\_\_\_\_\_\_\_\_ < 14 < \_\_\_\_\_\_\_\_\_$$

   b. Use your answers to Part A to indicate which two square roots $\sqrt{14}$ lies between.

   $$\_\_\_\_\_\_\_\_\_ < \sqrt{14} < \_\_\_\_\_\_\_\_\_$$

   c. Discuss with your partner which square root the value of $\sqrt{14}$ is closer to.

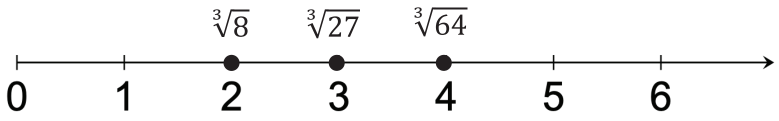
d. Plot $\sqrt{14}$ on the number line to indicate its approximate value based on your discussion.



Similar to the square roots of non-perfect squares, the cube roots of some numbers are irrational.

The values of such cube roots can be approximated using the cube roots of perfect cubes.

2. A number line is shown with values indicated at specific points.



a. Explain why $\sqrt[3]{8}$ is placed in the same location as 2 on the number line.

b. Label the locations of the following cube roots on the number line.

$\sqrt[3]{1}$, $\sqrt[3]{125}$, and $\sqrt[3]{0}$

c. Plot and label the approximate locations of the following cube roots on the number line.

$\sqrt[3]{9}$, $\sqrt[3]{10}$, and $\sqrt[3]{140}$

d. Compare your answers to your partner's. Discuss any differences and make revisions to your work as needed.

2.4.5 Guided Instruction: Approximating Irrational Numbers and Locating Them on a Number Line

The values of irrational numbers and expressions involving irrational numbers can be approximated.

1. Consider the expression $\sqrt{13}$.

   a. Between which two whole numbers does $\sqrt{13}$ lie on a number line?

   b. Which whole number is $\sqrt{13}$ is closer to?

   c. Faith thinks the value of $\sqrt{13}$ will be close to 3.6 or 3.7. Discuss with your partner why Faith may have made these estimates.

   d. Check Faith's estimates to determine if either estimate is close to the value of $\sqrt{13}$.

   e. Based on your work in Part D, complete the statement.

   The approximate value of $\sqrt{13}$, to the nearest tenth, is ____.____.

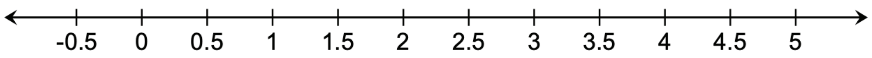
   f. Plot and label $\sqrt{13}$ on the number line to indicate its approximate location.

2. Use the approximation of $\sqrt{13}$ from the previous question to complete the following.

   a. Approximate the value of each expression.

Expression	Approximate Value
$\sqrt{13}+1$	
$\frac{\sqrt{13}}{2}$	
$4-\sqrt{13}$	

   b. Plot and label the approximate location of each expression from part a on the number line.



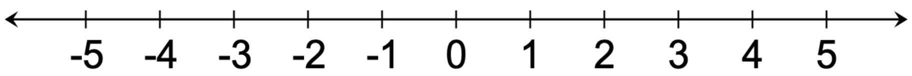
To approximate the value of expressions involving irrational numbers, first approximate the value of the irrational number. Then, use that approximation to perform any operations indicated.

2.4.6 Guided Practice: Approximating Irrational Numbers and Locating Them on a Number Line

1. Approximate the value of each expression.

Expression	Approximate Value
$-\pi$	
$\sqrt{6}$	
$\sqrt[3]{20}$	
$2 \cdot \sqrt{6}$	
$\pi - 3$	

2. Plot and label the approximate location of each expression from the previous question on the number line.



2.4.7 Wrap-Up: Cool Down

1. Explain the process for approximating an irrational number. Include each step in your process.